

COARSE-TO-FINE GAUSSIAN MOTION WITH SEMANTIC INITIALIZATION FOR TALKING HEAD SYNTHESIS UNDER UNSEEN AUDIO

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ABSTRACT

Audio-driven talking head models reconstruct appearance well but struggle with lip synchronization under unseen speech. Short training videos provide limited audio-motion coverage, while image supervision does not explicitly specify which Gaussians should share motion. FLAME-based approaches provide structure through mesh binding, but their audio-driven animation depends on the accuracy of intermediate facial-motion predictions. We propose a coarse-to-fine Gaussian motion representation with semantic initialization that directly incorporates motion sharing. Shared group motion encourages regional coordination, while per-Gaussian residuals preserve local articulation. External face parsing initializes soft group assignments, which are refined during early joint training with reconstruction and semantic supervision. These assignments organize shared motion while preserving differences among the lips, teeth, and mouth interior. Across four identities, our method achieves the best synchronization performance among the compared methods on two datasets. These gains on unseen audio are achieved while maintaining competitive self-reconstruction quality.

Index Terms— Talking head synthesis, out-of-distribution audio driven, synchronization robustness, semantic initialization

1. INTRODUCTION

Audio-driven talking head synthesis animates a person’s face from speech. Neural radiance field (NeRF) methods such as AD-NeRF, ER-NeRF, and RAD-NeRF have improved image quality, rendering efficiency and synchronization [1, 2, 3, 4]. More recently, 3D Gaussian splatting (3DGS) has provided an explicit representation for efficient rendering [5]. TalkingGaussian addresses motion inconsistency through persistent Gaussians and separate face and inner-mouth branches [6]. GaussianTalker focuses on efficient synthesis, while In-sTaG learns personalized models from short videos [7, 8]. Despite this progress, matching mouth movements to unseen speech remains difficult when training data are limited.

One reason is that a few minutes of video contain only a limited range of speech and mouth movements. A 3DGS model must learn how to deform many Gaussians from these frames, mainly by comparing rendered images with reference ones. This supervision does not directly specify how the Gaussians should move together. Shared networks and spatial features connect their predictions, but a good reconstruction of training images does not ensure coordinated motion under new speech. This matters particularly for actions such as lip closure and rapid opening and closing, which require several parts of the mouth to move together.

Explicit facial structure provides a useful way to guide Gaussian motion [9, 10, 11]. GaussianSpeech [12] binds Gaussians to a FLAME [13] mesh and predicts its deformation from speech, while GaussianHeadTalk [14] learns an audio-to-FLAME mapping to drive

a Gaussian avatar. These methods use an intermediate facial-motion prediction stage to connect audio with Gaussian deformation. We instead aim to incorporate facial structure directly into Gaussian motion learning by grouping Gaussians according to facial regions. This allows us to model shared regional motion without an additional audio-to-FLAME module.

In this paper, we propose a coarse-to-fine Gaussian motion representation for audio-driven talking head synthesis. The coarse level predicts shared deformation for groups of Gaussians, so the motion of each group is learned from multiple Gaussians together. The fine level adds a residual correction to each Gaussian to capture local differences. This combination supports coordinated movement while retaining the flexibility needed for detailed mouth motion, without binding Gaussians to a driving mesh.

To initialize the groups, we use external face parsing to identify coarse facial regions. Following the idea of incorporating segmentation cues into the Gaussian representation [15], we assign each Gaussian soft weights that determine how much it follows each group’s motion. Because a semantic region may contain different motions, these weights are updated during early joint training. Reconstruction supervision adjusts motion sharing, while semantic supervision keeps the grouping guided by facial regions. The groups therefore provide a starting structure without forcing every Gaussian in a region to follow one rigid motion.

Our contributions are summarized as follows: (1) We propose a coarse-to-fine motion representation that combines shared group deformation with local Gaussian corrections to learn coordinated facial motion from limited video. (2) We initialize soft motion groups using external face parsing and refine their assignments during early joint training, allowing the model to learn which Gaussians should share motion. (3) On four identities, our method improves average synchronization on unseen audio from CP-benchmark and THQA [16]. Self-reconstruction PSNR remains competitive with the evaluated methods.

2. METHOD

As shown in Figure 1, facial parsing initializes soft motion groups, whose assignments are refined during early joint training. Given audio features, a group network predicts shared motion and attribute offsets, which are blended for each Gaussian using its learned assignments. A residual network captures local articulation that shared motion alone cannot explain. The combined codes update Gaussian geometry and attributes, followed by rotation-aware rendering.

2.1. External parsing as group initialization

The canonical portrait is $\mathcal{G}^c = \{G_i^c\}_{i=1}^N$, where each Gaussian stores

$$G_i^c = \{\mathbf{u}_i^c, \mathbf{r}_i^c, \mathbf{s}_i^c, \alpha_i^c, \mathbf{SH}_i^c, \mathbf{E}_i\}. \quad (1)$$

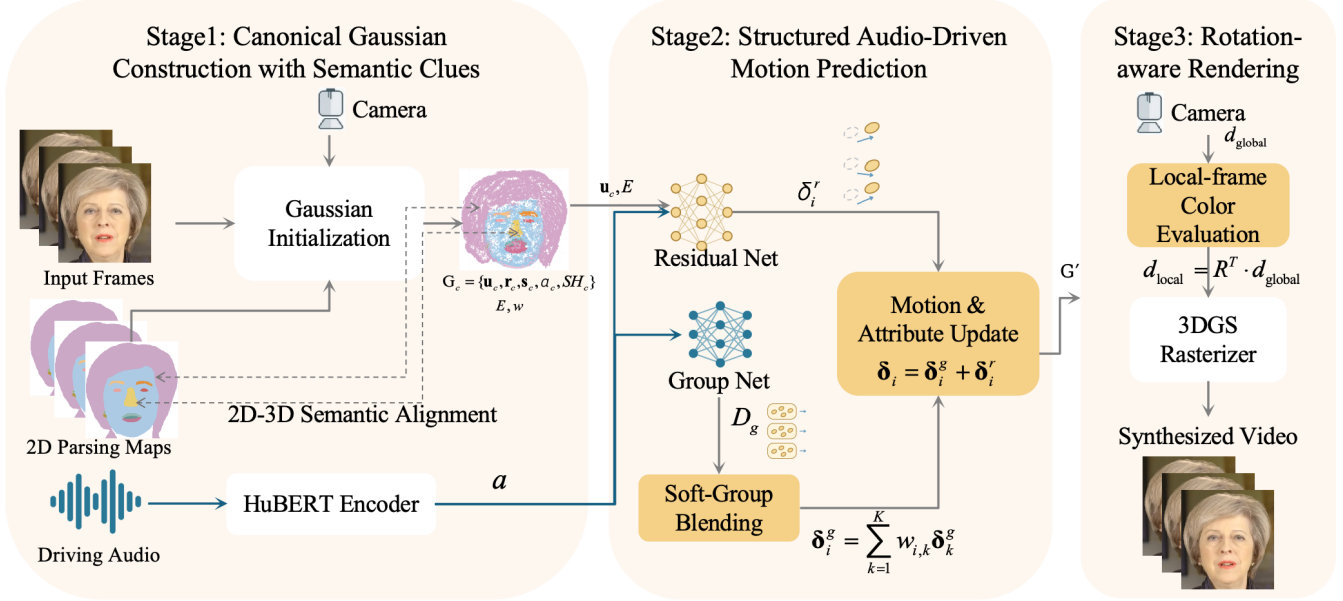


Fig. 1. Coarse-to-fine Gaussian motion. External parsing initializes Gaussian features and soft group weights. Audio-conditioned group codes provide shared deformation, and point-level residuals capture local corrections. The weights in “Soft Group Blending” are learned during early joint training. Group and residual codes update geometry and attributes before rendering.

These are the center, orientation, log-scales, opacity logit, spherical harmonics (SH) coefficients, and learnable semantic feature, respectively. We obtain coarse parsing as in TalkingGaussian [6] and refine lip and teeth labels using EasyPortrait [17]. We render Gaussian semantic features and use a lightweight classifier to predict per-pixel semantic probabilities. The 2D parsing maps supervise the rendered semantic predictions through the semantic loss defined in Section 2.4. Applying the same classifier to each Gaussian’s semantic feature yields soft assignments over K motion groups. At optimization iteration m , we write

$$\mathbf{w}_i^{(m)} = \text{softmax}\left(f_{\theta(m)}(\mathbf{E}_i^{(m)})\right), \quad \sum_k \mathbf{w}_{i,k}^{(m)} = 1. \quad (2)$$

Here f_{θ} is the classifier with parameters θ . The semantic class indices name the initial groups. Subsequent weights are motion-blending coefficients, and need not remain calibrated semantic probabilities. Soft weights allow multiple groups to influence a Gaussian, including at uncertain boundaries.

2.2. Audio-conditioned shared and residual deformation

Facial articulation involves both coordinated regional motion and local differences. We therefore combine shared group motion with per-Gaussian residuals to encourage regional coordination while retaining flexibility for local articulation.

Let $\mathbf{a}_t$ denote the HuBERT feature [18] for frame t . The group network predicts K shared motion and attribute codes:

$$\mathbf{D}_t^g = \Phi_{\text{group}}(\mathbf{a}_t) \in \mathbb{R}^{K \times 10}. \quad (3)$$

Each row is $\delta_{k,t}^g = [\mathbf{v}_{k,t}, \omega_{k,t}, \Delta\alpha_{k,t}, \Delta\mathbf{s}_{k,t}]$. Here $\mathbf{v}_{k,t} \in \mathbb{R}^3$ is a translation code and $\omega_{k,t} \in \mathbb{R}^3$ is a rotation vector, with $[\omega_{k,t}]_{\times} \in \mathfrak{so}(3)$. $\Delta\alpha_{k,t} \in \mathbb{R}$ and $\Delta\mathbf{s}_{k,t} \in \mathbb{R}^3$ are opacity and log-scale offsets. We describe the ten active output channels, omitting the three

unused color-offset channels in the implementation. The rotation-and-translation (RT) parameterization is motivated by rigid-motion representations in dynamic scene reconstruction [19, 20].

For each Gaussian, we blend the group codes using its learned soft assignments, following the general idea of weighted local deformation [21, 22]. A residual branch adds local corrections. Suppressing the training index m , the final code is

$$\delta_{i,t}^g = \sum_k \mathbf{w}_{i,k} \delta_{k,t}^g, \quad (4)$$

$$\delta_{i,t}^r = \lambda_{\text{res}} \Phi_{\text{res}}(\gamma(\mathbf{u}_i^c), \mathbf{a}_t, \mathbf{E}_i, \mathbf{e}_t), \quad (5)$$

$$\delta_{i,t} = \delta_{i,t}^g + \delta_{i,t}^r. \quad (6)$$

Here γ encodes canonical position, $\mathbf{e}_t$ denotes the auxiliary expression condition, and λ_{res} scales the residual corrections. Both branches are optimized jointly using the training objective in Section 2.4.

2.3. Local deformation and rendering

For the final rotation code, $\mathbf{R}_{i,t} = \exp([\omega_{i,t}]_{\times}) \in SO(3)$, where $[\cdot]_{\times}$ forms a skew-symmetric matrix. We introduce semantic-group pivots to avoid rotation around the global origin:

$$\mathbf{o}_k = \text{sg}\left(\frac{\sum_i \mathbf{w}_{i,k} \mathbf{u}_i^c}{\sum_i \mathbf{w}_{i,k} + \epsilon}\right), \quad \mathbf{o}_i = \text{sg}\left(\sum_k \mathbf{w}_{i,k} \mathbf{o}_k\right). \quad (7)$$

Here sg denotes stop-gradient and ϵ ensures numerical stability. With translation scale η_u , the effective translation and the deformed center are

$$\begin{aligned} \Delta \mathbf{u}_{i,t,\text{eff}} &= \eta_u \mathbf{v}_{i,t} + \mathbf{o}_i - \mathbf{R}_{i,t} \mathbf{o}_i, \\ \mathbf{u}_{i,t} &= \mathbf{R}_{i,t} (\mathbf{u}_i^c - \mathbf{o}_i) + \mathbf{o}_i + \eta_u \mathbf{v}_{i,t}. \end{aligned} \quad (8)$$

Let Q convert rotation matrices to quaternions. The updated orientation is

$$\mathbf{r}_{i,t}^c = \text{norm}(Q(\mathbf{R}_{i,t}) \otimes \mathbf{r}_i^c), \quad (9)$$

Table 1. Self-reconstruction and synchronization under unseen driving audio. Bold marks the best value in each column.

Method	Self-reconstruction			CP-benchmark		THQA	
	PSNR $\uparrow$	Sync-D $\downarrow$	Sync-C $\uparrow$	Sync-D $\downarrow$	Sync-C $\uparrow$	Sync-D $\downarrow$	Sync-C $\uparrow$
ER-NeRF	28.406	8.802	6.118	9.722	4.992	9.736	3.550
TalkingGaussian	32.495	8.112	6.918	8.840	5.660	9.529	3.655
GaussianTalker	32.802	8.643	6.761	9.440	5.662	9.891	3.807
InsTaG	30.255	11.075	3.575	10.880	3.524	10.706	2.422
InsTaG++	32.333	9.314	5.386	9.524	4.903	9.748	3.245
Ours	32.350	7.643	7.562	8.426	6.324	9.223	4.147

where $\otimes$ denotes quaternion multiplication and norm denotes ℓ_2 normalization. Scale and opacity offsets are applied before activation:

$$\mathbf{s}'_{i,t} = \exp(\mathbf{s}_i^c + \Delta \mathbf{s}_{i,t}), \quad \alpha'_{i,t} = \text{sigmoid}(\alpha_i^c + \Delta \alpha_{i,t}). \quad (10)$$

The rotation code is blended before the exponential map. This is not a weighted average of rigid transformation matrices or a full exponential map on $SE(3)$. Soft blending and point residuals do not constrain an entire semantic region to one rigid transform.

We map the world-space viewing direction back to the canonical reference frame to account for the predicted rotation during view-dependent color evaluation:

$$\begin{aligned} \mathbf{d}_{i,t,\text{local}} &= \mathbf{R}_{i,t}^\top \mathbf{d}_{\text{global}}, \\ \mathbf{c}'_{i,t} &= \text{SH}(\text{SH}_i^c, \mathbf{d}_{i,t,\text{local}}). \end{aligned} \quad (11)$$

2.4. Joint reconstruction and semantic training

The semantic loss is defined as

$$\mathcal{L}_{\text{sem}} = -\frac{1}{|\Omega| \log K} \sum_{\mathbf{p}, k} c_k y_k(\mathbf{p}) \log \hat{p}_k(\mathbf{p}). \quad (12)$$

Here Ω is the pixel set; c_k , y_k , and $\hat{p}_k$ denote class weights, target labels, and predicted probabilities.

Training combines image, semantic, and auxiliary losses:

$$\begin{aligned} \mathcal{L}_{\text{img}} &= \mathcal{L}_1 + \lambda_s \mathcal{L}_{\text{SSIM}} + \mathcal{L}_{\text{perc}}, \\ \mathcal{L} &= \mathcal{L}_{\text{img}} + \lambda_{\text{sem}} \mathcal{L}_{\text{sem}} + \mathcal{L}_{\text{aux}}. \end{aligned} \quad (13)$$

Here $\mathcal{L}_1$ is mean absolute RGB error, and $\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(\hat{\mathbf{I}}_t, \mathbf{I}_t)$. $\mathcal{L}_{\text{perc}}$ uses weighted mouth-region or patch LPIPS [23]. $\mathcal{L}_{\text{aux}}$ collects weighted mouth L_1 and mean-squared errors, opacity and log-scale offset penalties, and head-mask and attention terms. Losses follow the training schedule; semantic supervision is periodic.

3. EXPERIMENTS

3.1. Evaluation protocol

We evaluate four identities: May, Lieu, Obama, and Shaheen. Each source video lasts 3–5 minutes at 25 frames per second. Frames are cropped to 512×512 , except for Obama, whose crop size is 450×450 . Self-reconstruction uses audio and camera trajectories from the held-out portion of the target sequence, with paired reference frames.

Unseen-audio evaluation uses CP-benchmark and audio from THQA [16]. CP-benchmark is a stress test set constructed using text constraints and text-to-speech (TTS) audio cloning. It crosses 12 articulation-stress prompts with four reference-prosody conditions per identity. The prompts include rapid alternating syllables, repetitive sequences, and short phrases that test lip closure. They vary the

Table 2. Ablation study. Bold indicates the best value in each column.

Variant	CP-benchmark		THQA	
	Sync-D $\downarrow$	Sync-C $\uparrow$	Sync-D $\downarrow$	Sync-C $\uparrow$
w/ point-wise offsets	8.571	6.224	9.349	4.060
w/o group RT	8.490	6.271	9.273	4.118
w/o point-wise RT	8.794	5.914	9.357	3.989
Full model	8.426	6.324	9.223	4.147

density and phonetic context of bilabial consonants /b/, /p/, and /m/. Each identity has 48 clips, totaling 192 clips and approximately 28 minutes across four identities. THQA evaluation uses 94 audio clips with a total duration of approximately 9.93 minutes. The same 94 clips drive each of the four identities. Neither set has paired visual ground truth.

We train a separate model for each identity using HuBERT features and $K = 21$ semantic motion groups. The Gaussian representation uses second-order SH coefficients. Soft group assignments are learned during early joint training and fixed after 30,000 iterations. We train each model for a total of 85,000 iterations.

We compare our method with ER-NeRF, TalkingGaussian, GaussianTalker, InsTaG and InsTaG++. InsTaG follows its original experimental setting, while InsTaG++ is trained on the full target-identity training data.

For unseen-audio synthesis, the auxiliary condition $\mathbf{e}_t$ contains facial action-unit (AU) features and an eye-blink control. These conditions and the camera trajectories are taken from the target identity’s training sequence. Reported metrics are averaged equally across the four identities.

Peak signal-to-noise ratio (PSNR) measures self-reconstruction fidelity against paired references. SyncNet [24] provides synchronization distance (Sync-D, lower is better) and synchronization confidence (Sync-C, higher is better) on all three evaluation sets.

3.2. Synchronization under unseen driving audio

Table 1 presents reconstruction fidelity alongside synchronization under unseen audio. Our model achieves a self-reconstruction PSNR of 32.350 dB, compared with 32.802 dB for GaussianTalker. It also achieves the best self-reconstruction Sync-D and Sync-C. Reconstruction fidelity provides a reference for interpreting the unseen-audio results. The main comparison focuses on synchronization when the driving speech changes.

3.3. Ablation Study

Table 2 compares different motion configurations. The *w/ point-wise offsets* variant directly predicts per-Gaussian position, quaternion,



Fig. 2. Word-level qualitative comparison under unseen CP-benchmark audio.

scale, and opacity offsets. The *w/o group RT* variant uses point-level codes for translation, rotation, and scale, while retaining both branches for opacity. The *w/o point-wise RT* variant uses group-level translation and rotation, while retaining both branches for scale and opacity. Semantic-group pivots are used in *w/o point-wise RT* and the full model, whereas *w/o group RT* applies rotations about the canonical origin.

The full model achieves the best synchronization scores among these configurations on both audio sets. The *w/o point-wise RT* variant performs worse despite retaining local scale and opacity corrections, supporting the use of point-level RT refinement alongside shared group motion.

3.4. Qualitative comparison

Figure 2 compares mouth motion in two word-level examples from unseen CP-benchmark audio. For May’s “Rob”, our model shows a smaller mouth opening at 0.28 s than InsTaG. For Shaheen’s “bam-

boo”, our model captures the opening–closure transition from 1.48 s to 1.56 s, a pattern also observed in TalkingGaussian. InsTaG retains a larger opening at 1.56 s, whereas InsTaG++ shows a weaker transition between these frames.

4. CONCLUSION

We propose a coarse-to-fine Gaussian motion representation with semantic initialization for synchronization under unseen speech. Group-level shared deformation and point-level residual deformation provide regional coordination and local articulation under limited supervision. Soft group assignments are learned during early joint training. Across four identities, the full model achieves the best average Sync-D and Sync-C among the compared methods on CP-benchmark and THQA. It also maintains competitive self-reconstruction PSNR. These results support synchronization generalization within the evaluated setting.

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